



Council for Technical Education and Vocational Training

Office of the Controller of Examinations

Sanathimi, Bhaktapur

Regular/Back/Scholarship Exam – 2081/2082 Chaitra/Baishakh

Program: Diploma in Computer Engineering/
Information Technology

Full Marks: 80

Year/Part: II/I (2022) © Arjun

Pass Marks: 32

Subject: Basics Electrical and Electronics
Engineering

Time: 3 hrs.

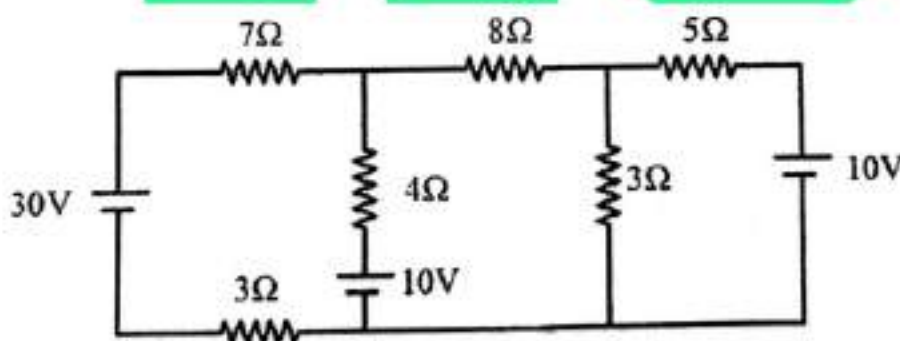
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.



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Attempt any **FIVE** questions.

1. a. Derive current flow in a circuit with necessary diagram. [6+2]
Also, write about electromotive force.
- b. Define passive component. Explain resistor in a series [1+3]
connection.
- c. Explain Kirchhoff's current law with suitable example. [4]
2. a. Define the following terms: [4×1]
 - i. Time period
 - ii. Frequency
 - iii. Amplitude
 - iv. Phase
- b. Use nodal analysis to find current through 8Ω resistor in [8]
the following figure below:



- c. Differentiate between active power and reactive power. [4]
3. a. Derive the expression for the rms and average value of [8]
sinusoidal current and voltage.

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Cont.

- b. A 230V, 50 Hz AC supply is applied a coil of 0.05H inductance and 10Ω resistance connected in series with a capacitor of $20\mu\text{F}$. Calculate the following: [8]
- i. Impedance
 - ii. Power factor
 - iii. Circuit current
 - iv. Active power
4. a. Define semiconductor. Explain its types in detail. [2+6]
- b. Explain how zener diode works as voltage regulator with clear diagram and expression. [8]
5. a. Define transistor. Explain the input characteristics of CE configuration with necessary diagram. [1+7]
- b. What are the characteristics of OP-AMP? Explain summing amplifier. [3+5]
6. Write short notes on: (any **FOUR**) [4×4]
- a. Thevenin's theorem
 - b. BJT as switch
 - c. MOSFET
 - d. CMOS
 - e. Clamper circuit
 - f. Half wave rectifier

Good Luck !



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AC



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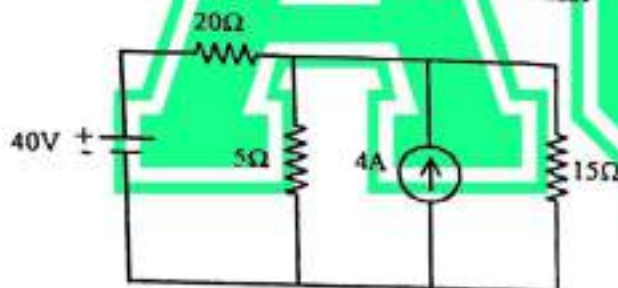
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Attempt any **EIGHT** questions.

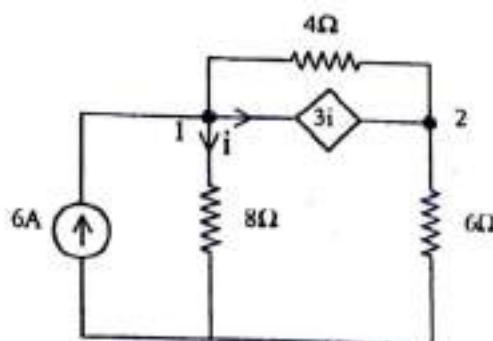


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1. a. State Ohm's law. Explain different types of voltage and current source. [5]
b. The coil of a relay takes a current of 0.12 A at temperature of 15°C and connected across a 60 V supply. If the minimum operating current of the relay is 0.1 A, calculate the temperature above which the relay will fail to operate when connected to the same supply. (Resistance temperature coefficient of coil material is $0.0043^{\circ}\text{C}^{-1}$ at 6°C). [5]
2. a. State and prove the maximum power transfer theorem. [5]
b. State Thevenin's theorem. Find current through 15Ω resistor using Thevenin's theorem. [5]



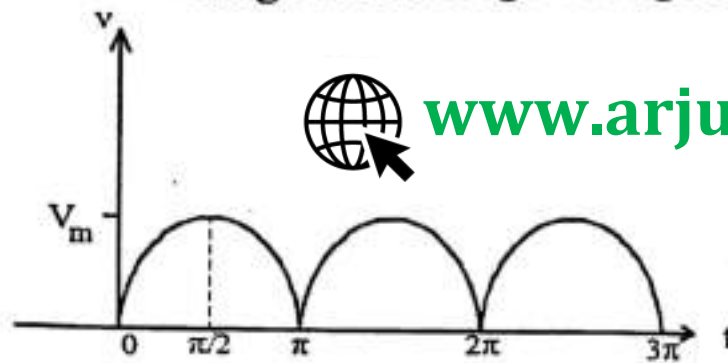
3. a. Using nodal analysis find value of i . [5]



Cont.

b. Derive power for pure resistance and prove that: [5]
 $P_{avg} = V_{rms} \cdot I_{rms}$.

4. a. Find rms voltage and average voltage of given figure. [5]



b. A sinusoidal voltage is applied to three parallel branches yielding branch currents $i_1 = 14.14 \sin(\omega t - 45^\circ)$, $i_2 = 28.3 \cos(\omega t - 60^\circ)$ and $i_3 = 7.07 \sin(\omega t + 60^\circ)$. Find complete time expression for source current and draw the phasor diagram using voltage as reference. [5]

5. a. State the importance of power factor. [5]

b. A 220V, 50Hz circuit consists of 500Ω resistor, inductance of 2H and capacitor of $10\mu F$ in parallel. Find: [5]
(i) total current (ii) complex power (iii) active power
(iv) reactive power (v) power factor.

6. a. Show the intrinsic and extrinsic characteristics of P-type semiconductor. [5]

b. What is rectifier used for? Explain full wave rectifier. [5]

7. a. Compare common emitter and common collector configuration.

b. How BJT can be used as logic gates? Explain. [5]

8. a. Show the working principle of CMOS. [5]

b. Write in detail about inverting and non-inverting mode amplifier. [5]

9. Write short notes on: (any **TWO**) [2×5]

a. Types of MOSFET

b. Zener diode

c. Kirchhoff's law

d. Differentiator

Good Luck !

AC

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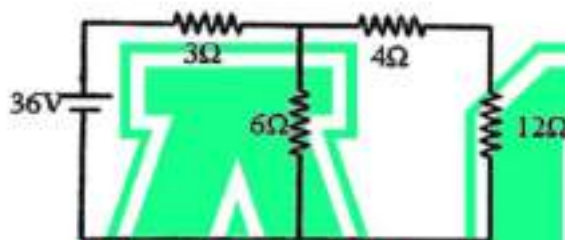
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Attempt Any Five questions.



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1. a) Define electromotive Force and potential difference. [4]
b) Explain Kirchhoff's voltage law with suitable example. [4]
c) Using Thevenin's theorem find the current through 12Ω resistor from below circuit. [8]



2. a) State and prove maximum power transfer theorem. [4]
b) Define the following forms. [4x2=8]
i) Instantaneous value ii) Average value
iii) Peak value iv) RMS value
c) Explain VCVS and CCVS. [4]
3. a) Derive the expression for the r.m.s. and average value of sinusoidal current and voltage. [8]
b) A 230V, 50Hz AC supply is applied a coil of 0.06H inductance and 5Ω resistance connected in series with a capacitor of $10\mu\text{F}$. Calculate the following. [4x2=8]
i) Impedance ii) Circuit current
iii) Power factor iv) Active power
4. a) What do you mean by rectifier? Explain the working principle of half-wave rectifier with waveform. [8]

- b) What do you mean by transistor? Explain BJT as a switch with neat diagram. [8]
5. a) Explain the construction and working of CMOS. [8]
b) What do you mean by operational amplifier? Explain the ideal and real characteristics of operational amplifier. [8]
6. Write short notes on : (Any Four) [4x4=16]
a) Norton's theorem b) Semiconductor diode
c) BJT as a logic gate d) MOSFET
e) Clipper circuit

Good Luck !



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